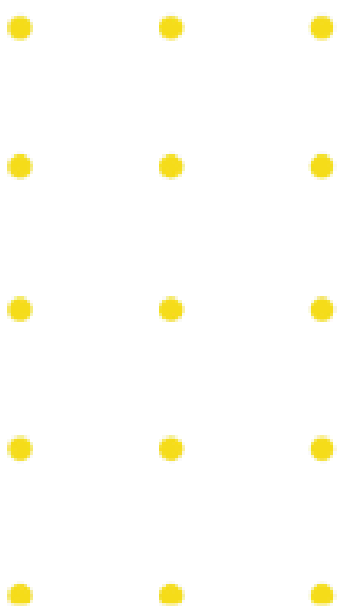


Module 10:

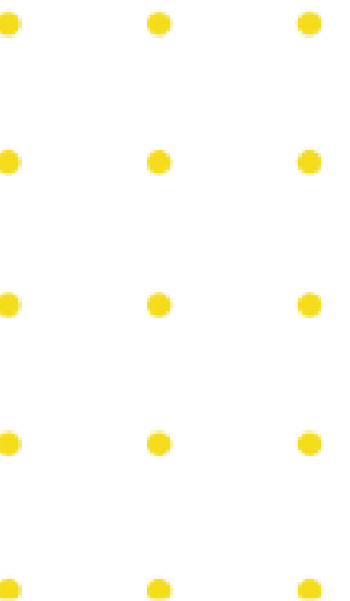
Digital Forensics &

Incident Response

DFIR



Topic 2: Digital Forensics



What is Digital Forensics

- ❑ Digital forensics is the process of collecting, preserving and analysing the acquired data to be presented as evidences against cyber criminals in court.
- ❑ Cyber criminals are individuals with high technical skills who use their skills to perform malicious activities known as cyber crimes against organizations for the purpose of financial, political, or self benefits.



Common Cyber Crime Types

- **Phishing and Scam:**
 - Type of social engineering attack that targets the user and tricks them by sending fake messages and emails.
- **Identity Theft:**
 - Using another person's personal data like credit card numbers or personal pictures without their permission to commit a fraud.
- **Ransomware Attacks:**
 - It is a type of malware that has the capability to prevent users from accessing all of their personal data on the system by encrypting them and then asking for a ransom.
- **Internet Fraud:**
 - Makes use of the internet and it can be considered a general term that groups all of the crimes that happen over the internet like spam, banking frauds, theft of service, etc.

Digital Forensics Types

Web Forensics

 Web log analysis.

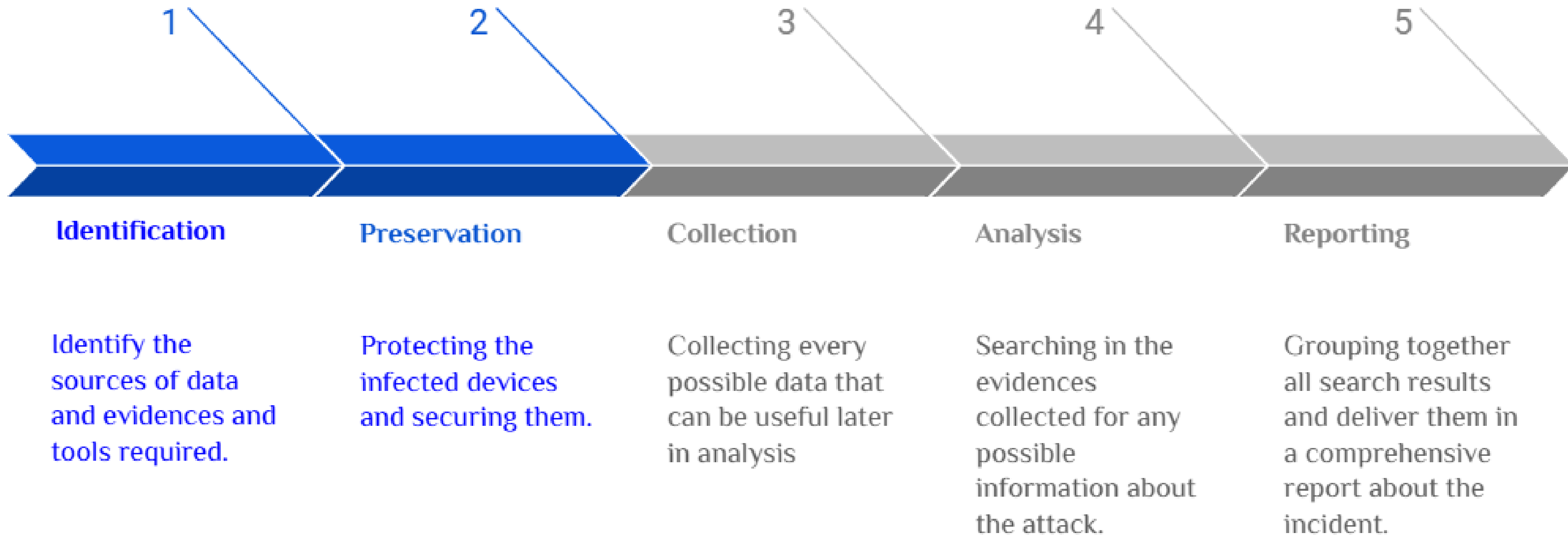
Network Forensics

 Network traffic analysis

Computer Forensics

 Disk Forensics
Memory Forensics
Browser Forensics

Digital Forensics Process



Computer Forensics: Disc Forensics

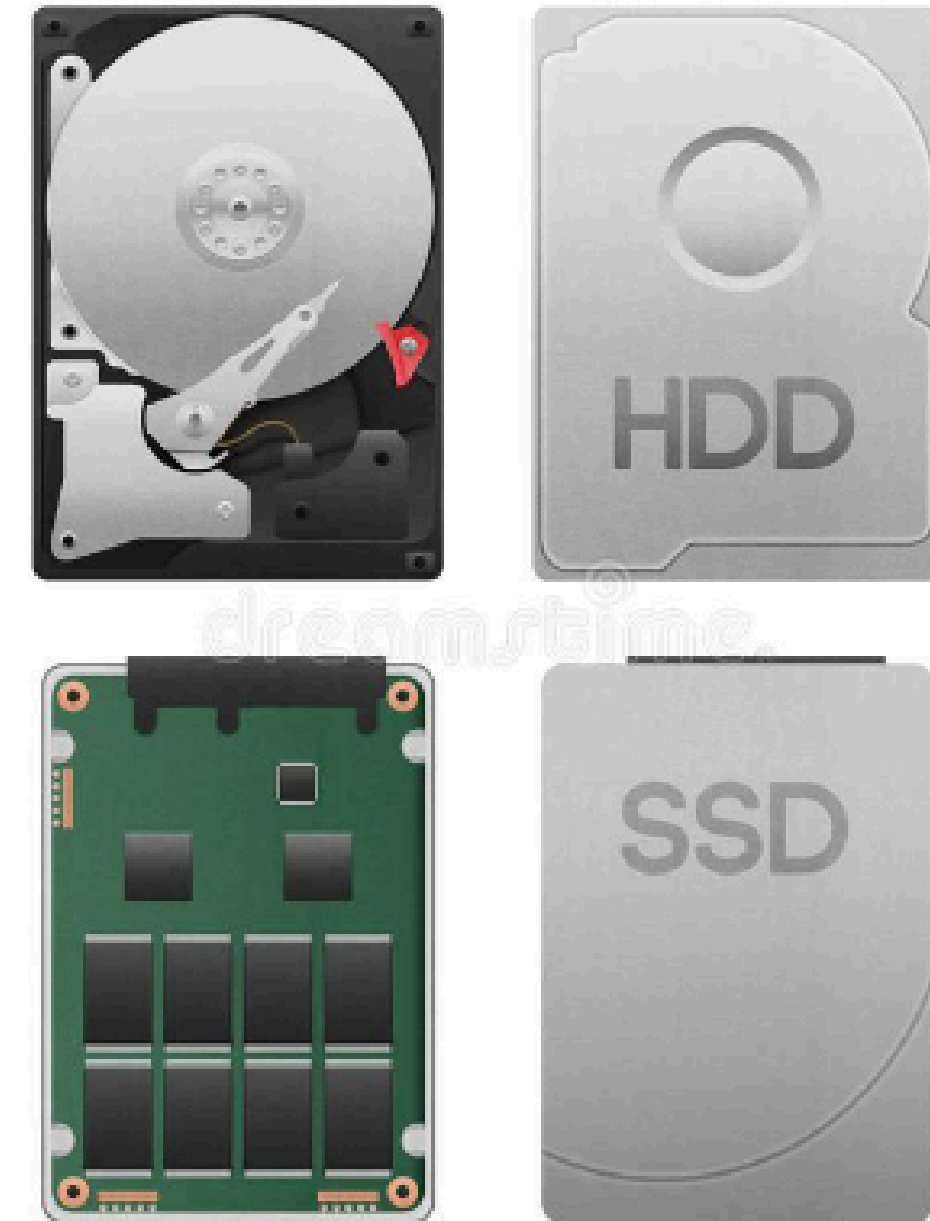
Hard Disk Types:

- HDD (Hard Disk Drivers):

They are a very old type of storage device and it's been around for over 50 years, mainly relying on spinning disks or platters to read and write data.

- SSD (Solid State Drivers):

A new type of storage device and it uses flash memory techniques to improve performance and they are smaller, faster and more reliable than HDD's.



File System Types

NTFS:

- Also known as NT file system is a property journaling file system developed by microsoft starting from Windows NT 3.1, it was the replacement of FAT (File Allocation Table) filesystem, NTFS holds powerful disk utilisation capabilities and can manage disk space more effectively.

FAT32:

- Also known as File Allocation Table, It is a better version of FAT16 that includes a better compatibility with different devices, it was employed on old windows versions but now most USB devices use FAT32 as a filesystem.

EXT3, EXT4:

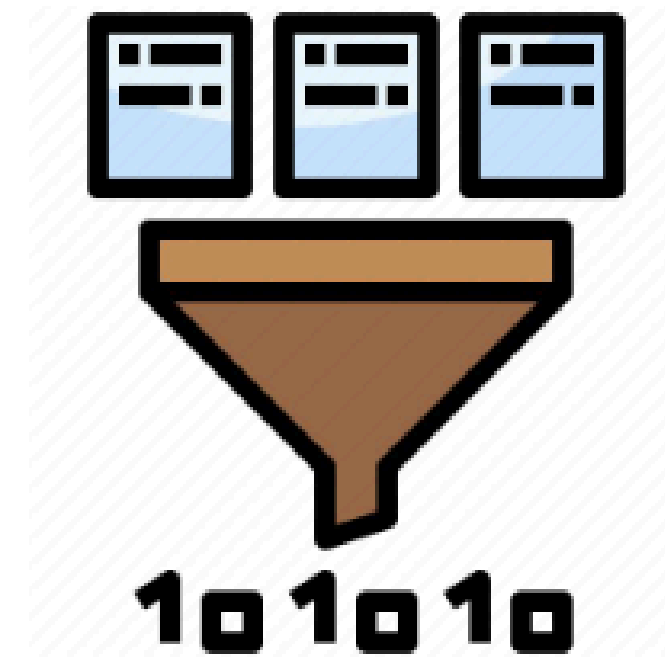
- Also known as Extended Filesystems are journaled file system commonly used by the Linux kernel, EXT4 is similar to EXT3 but has better performance and better file system support.

Hashing & Integrity

A Hash is a unique fingerprint of a fixed size that is produced by a function called Hash Function or a Hash Algorithm.

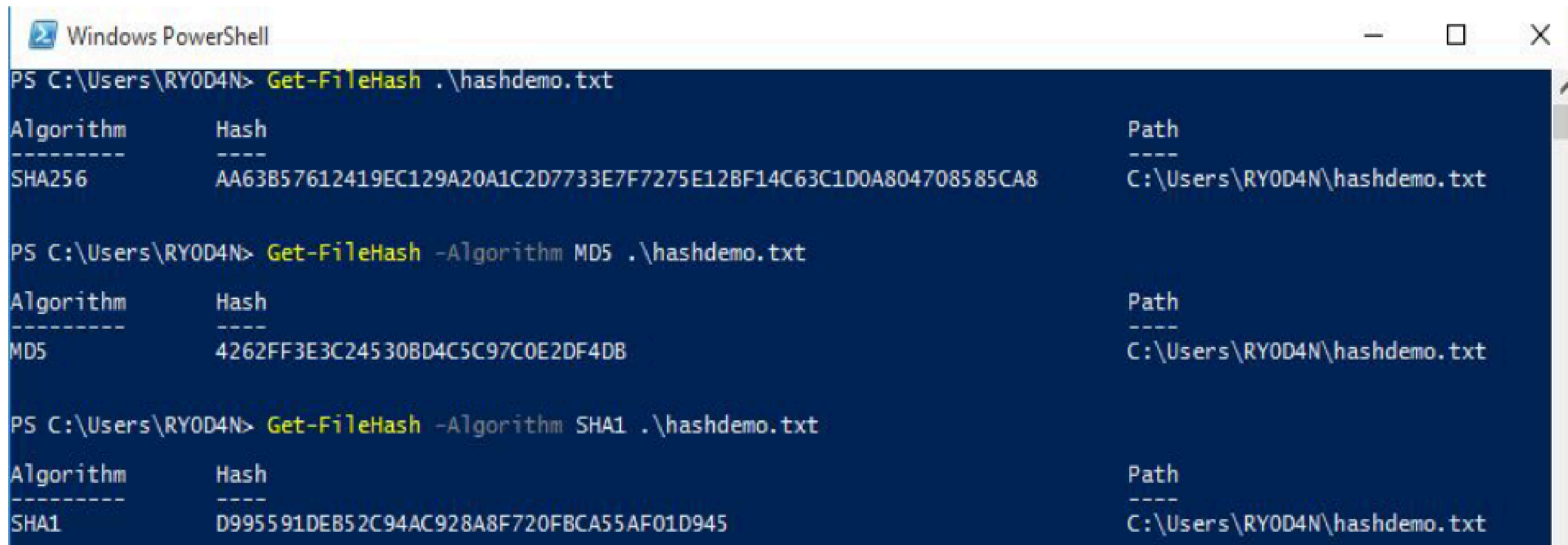
Common hashing Algorithms:

- MD5 (can produce collisions).
- SHA1 (can produce collisions).
- SHA256 (secure).
- SHA512 (more secure).



Hashing

- Use **Powershell** to calculate the hash of a file.
- Change any byte in the file.
- Re-hash the file again and observe the difference.



```
Windows PowerShell
PS C:\Users\RY0D4N> Get-FileHash .\hashdemo.txt

Algorithm      Hash
-----
SHA256         AA63B57612419EC129A20A1C2D7733E7F7275E12BF14C63C1D0A804708585CA8
C:\Users\RY0D4N\hashdemo.txt

PS C:\Users\RY0D4N> Get-FileHash -Algorithm MD5 .\hashdemo.txt

Algorithm      Hash
-----
MD5            4262FF3E3C24530BD4C5C97C0E2DF4DB
C:\Users\RY0D4N\hashdemo.txt

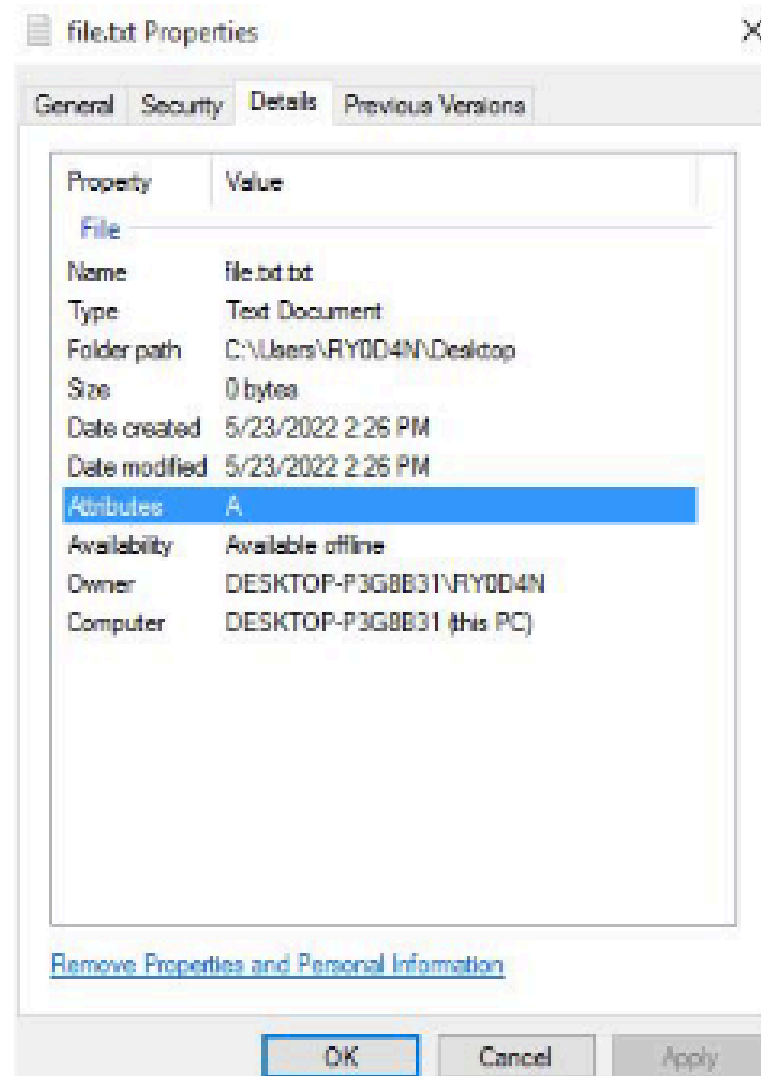
PS C:\Users\RY0D4N> Get-FileHash -Algorithm SHA1 .\hashdemo.txt

Algorithm      Hash
-----
SHA1           D995591DEB52C94AC928A8F720FBCA55AF01D945
C:\Users\RY0D4N\hashdemo.txt
```

MetaData

Metadata is : Data about data.

Metadata can be a great and easy to collect source of information about files.



```
[chorollo@parrot]~$ exiftool secret
ExifTool Version Number      : 12.04
File Name                    : secret
Directory                    : .
File Size                    : 128 bytes
File Modification Date/Time   : 2022:05:10 14:58:42-04:00
File Access Date/Time        : 2022:05:10 15:09:24-04:00
File Inode Change Date/Time   : 2022:05:10 15:09:23-04:00
File Permissions              : rw-r--r--
Error                        : Unknown file type
```

Disk Forensics

- **Disk forensics tools:**

- FTK Imager (disk acquisition).
- Kape (live forensics).
- Autopsy (disk image analysis).



Windows Disk imaging:

- Logical Images:

For logical drives only like C:/ D:/ F:/ and can't capture the unallocated space.

- Physical Images:

An Image for the whole physical drive and includes every piece of data.

FTK Imager

Installation:

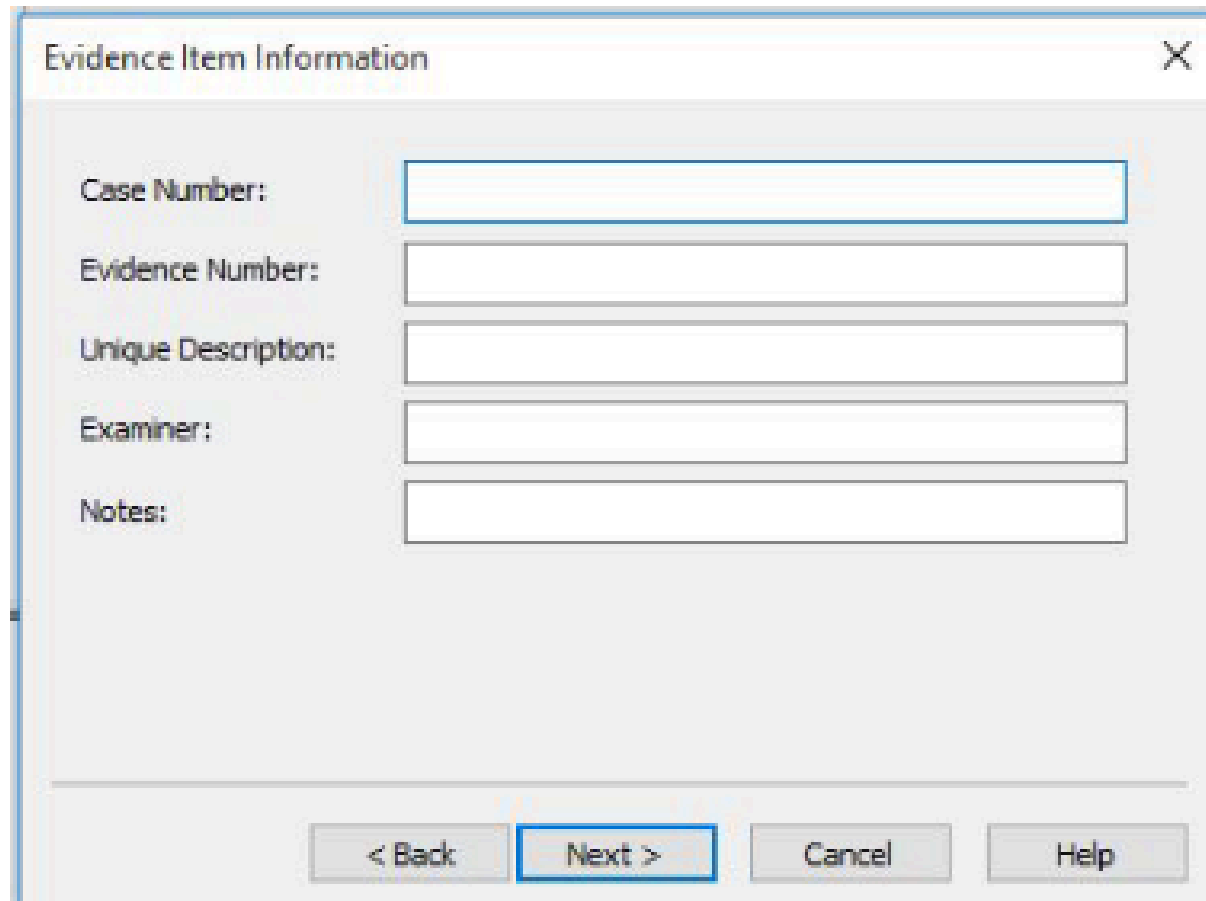
- Install the latest version from:
<https://accessdata.com/product-download/ftk-download-page>

FTK core features:

- Taking a forensics disk image
- Dumping memory (RAM).
- view disk images content.
- Generate hashes.
- Export files from disk images.

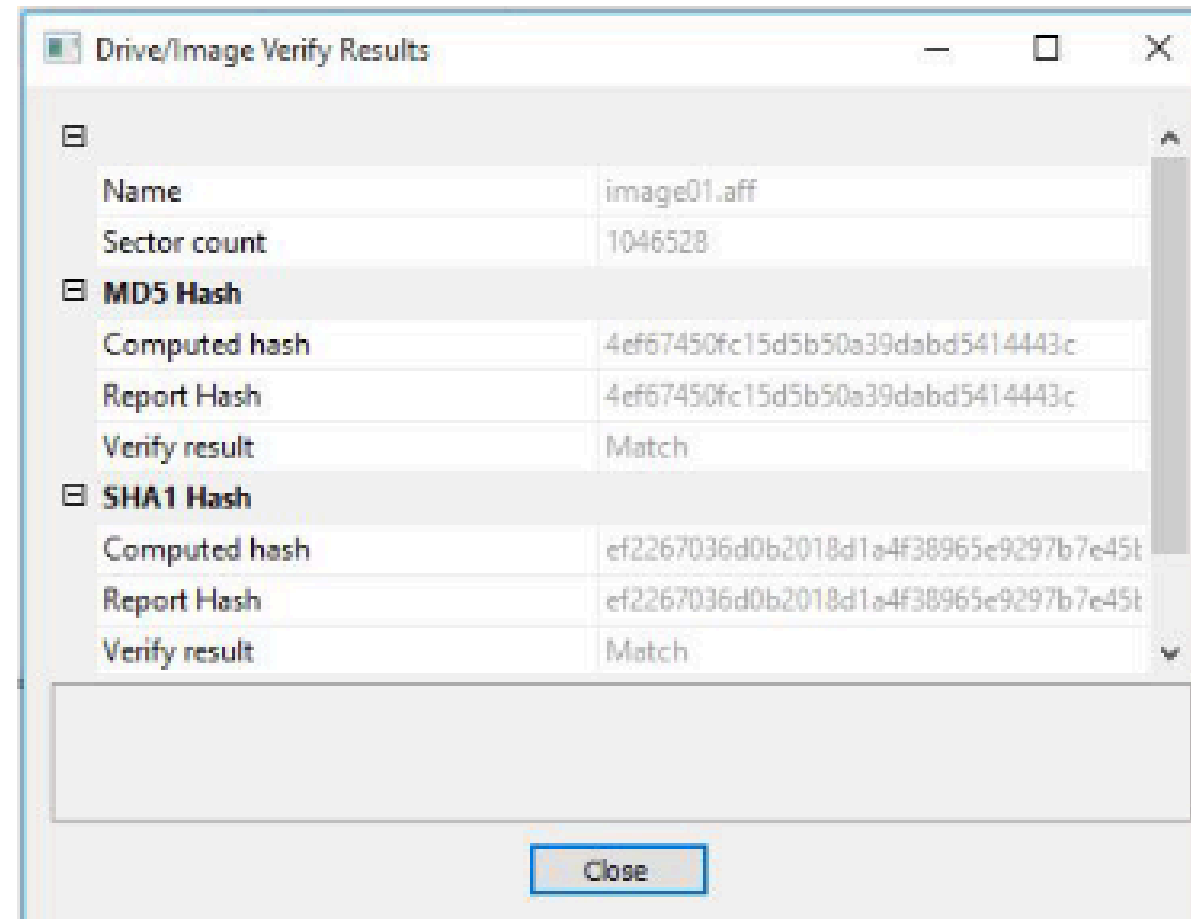
Disk Acquisition

- Use FTK Imager to acquire a disk image.
- Explore different image formats.
- Fill the evidence item information as it is important for the investigation integrity
- Check the hash values for the resulted image and verify the results.



The 'Evidence Item Information' dialog box is a standard Windows-style window with a title bar and a close button. It contains five input fields: 'Case Number:', 'Evidence Number:', 'Unique Description:', 'Examiner:', and 'Notes:'. Each field is represented by a text label followed by a rectangular input box. At the bottom of the dialog, there are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'. The 'Next >' button is highlighted with a blue border.

Field	Value
Case Number:	
Evidence Number:	
Unique Description:	
Examiner:	
Notes:	



The 'Drive/Image Verify Results' dialog box displays verification data for a file named 'image01.aff'. It shows the sector count and MD5/SHA1 hash results. The MD5 hash verification shows a 'Match' between the computed and report hashes. The SHA1 hash verification also shows a 'Match'. A 'Close' button is located at the bottom right.

Property	Value
Name	image01.aff
Sector count	1046528
MD5 Hash	
Computed hash	4ef67450fc15d5b50a39dabd5414443c
Report Hash	4ef67450fc15d5b50a39dabd5414443c
Verify result	Match
SHA1 Hash	
Computed hash	ef2267036d0b2018d1a4f38965e9297b7e45t
Report Hash	ef2267036d0b2018d1a4f38965e9297b7e45t
Verify result	Match

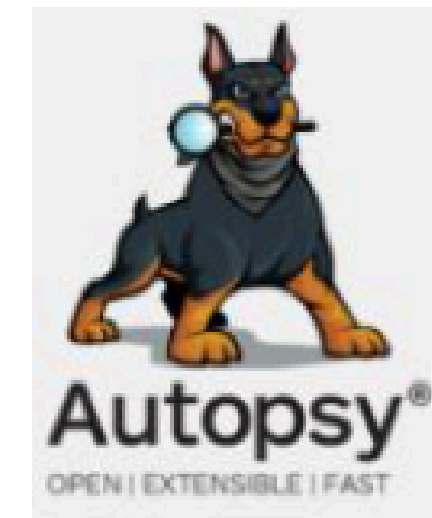
Disk Analysis: Autopsy

What is Autopsy:

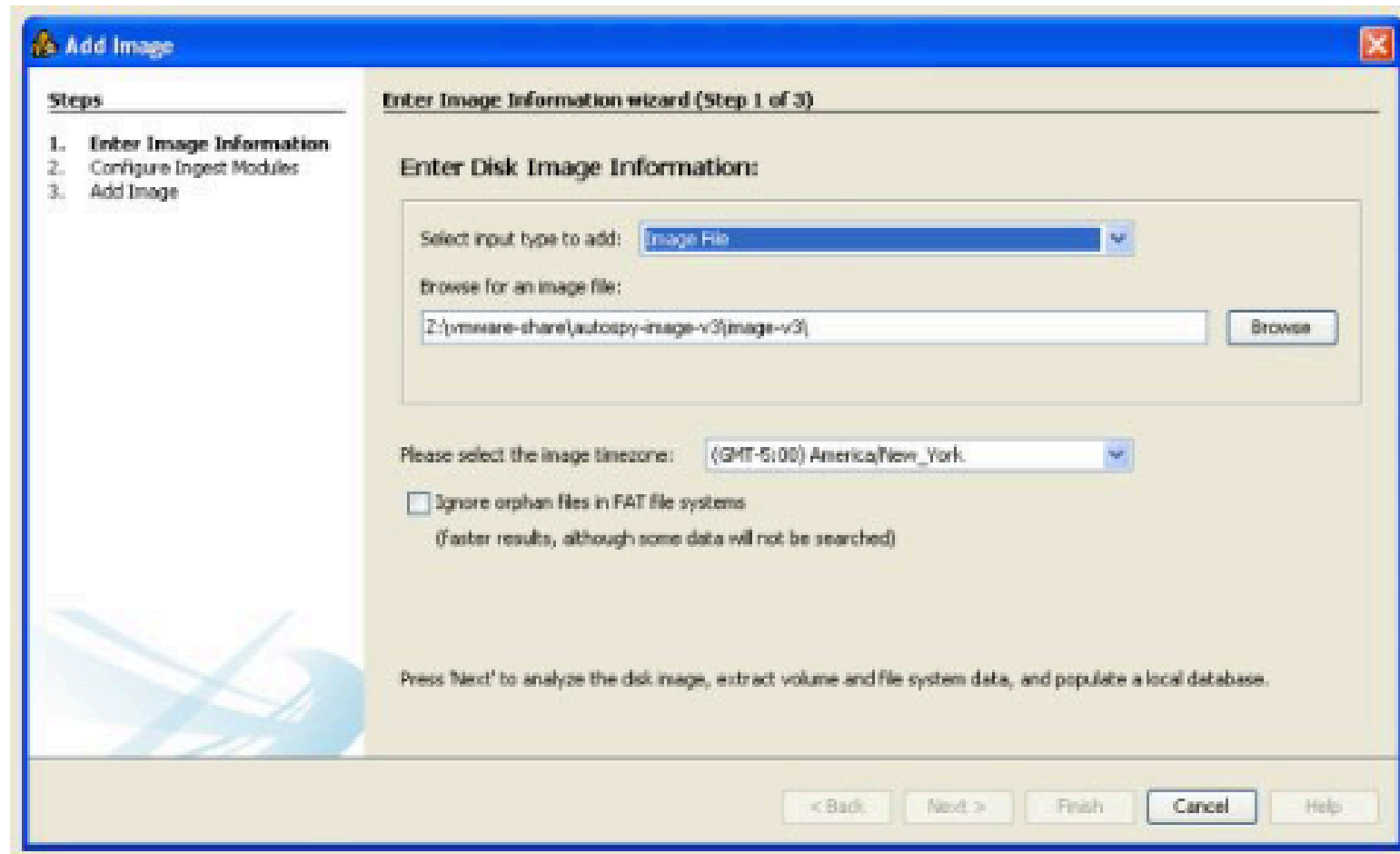
- Autopsy is a forensics tool that is used to investigate and analyse images from computer or mobile phones.

Autopsy Features:

- Cases can be shared between multi users.
- Support for multiple file systems (NTFS, FAT12/FAT16/FAT32/ExFAT, (CD-ROM), (Ext2/Ext3/Ext4).
- Ability to tag files such as “suspicious” tag and filter known good files.
- File type detection based on signatures and extension mismatch.
- Android Analysis: CALLS, SMS, etc..



Autopsy in Windows



Autopsy

Open autopsy and start a case then load a disk image and navigate through the windows to be familiar with the tool.

The screenshot displays the Autopsy application window. The top menu bar includes options like 'Add Data Source', 'Images/Videos', 'Communications', 'Timeline', 'File Discovery', 'Close Case', and 'Generate Report'. Below the menu, there's a toolbar with icons for navigation and search. The left sidebar shows a tree view of 'Data Sources' and 'Views'. Under 'Views', 'File Types' is expanded, showing categories like 'Images (1171)', 'Videos (36)', 'Audio (150)', 'Archives (228)', 'Databases (6)', 'Documents', 'Executable', and 'Deleted Files'. The 'Results' section on the left lists various extracted content items such as 'Encryption Suspected (2)', 'Extension Mismatch Detected (9)', 'Installed Programs (32)', 'Operating System Information (2)', 'Operating System User Account (8)', 'Recent Documents (8)', 'Shell Bags (51)', 'USB Device Attached (1)', 'Web Bookmarks (6)', 'Web Cookies (24)', 'Web History (887)', 'Web Search (4)', 'Keyword Hits', 'Single Literal Keyword Search (0)', 'Single Regular Expression Search (0)', 'Email Addresses (11275)', 'Hashset Hits', and 'E-Mail Messages'.

The main pane is titled 'Listing' and shows a table of files. The table has columns for 'Name', 'S', 'C', 'O', 'Modified Time', 'Change Time', 'Access Time', 'Created Time', 'Size', 'Flags(Dr)', and 'File'. The files listed are all PNG images, including 'gr4005.png', 'ia3002_1.png', 'ia3002_2.png', 'lqc202.png', 'lqc207.png', 'lqc209.png', 'lqc20a.png', 'lqc291.png', 'SV-262e1.png', 'SV-262e3.png', 'SV-262e4.png', and 'sv2611.png'. The 'lqc209.png' file is currently selected.

Below the table, there are tabs for 'Hex', 'Text', 'Application', 'Message', 'File Metadata', 'Context', 'Results', 'Annotations', and 'Other Occurrences'. The 'Text' tab is active, showing a preview of the selected file. The preview area displays a black video game controller.

Computer Forensics: Memory Forensics

Memory Acquisition:

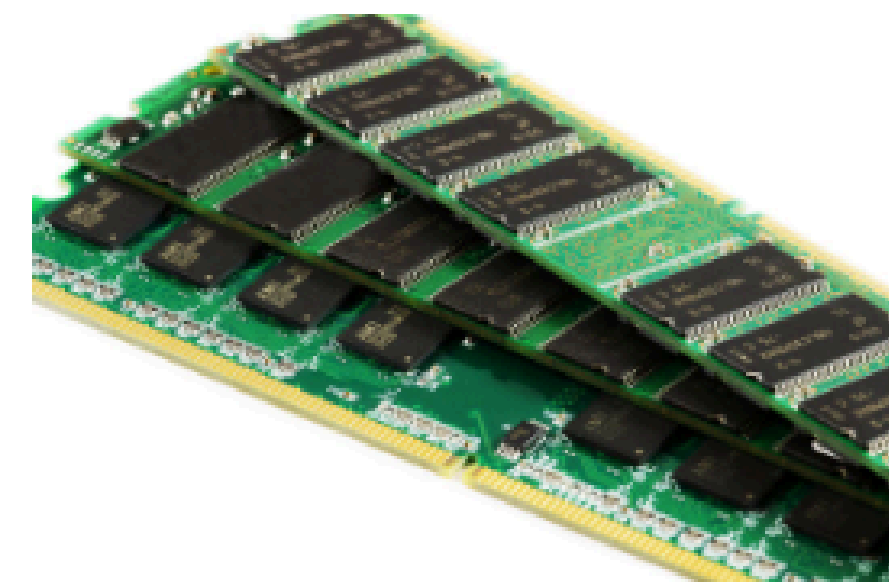
- Memory acquisition is the process of copying the whole content of the physical memory to another storage device in order to further investigate it.

Memory Acquisition tools:

- FTK Imager.
- DumpIt.
- Linux Memory Extractor (LiME).

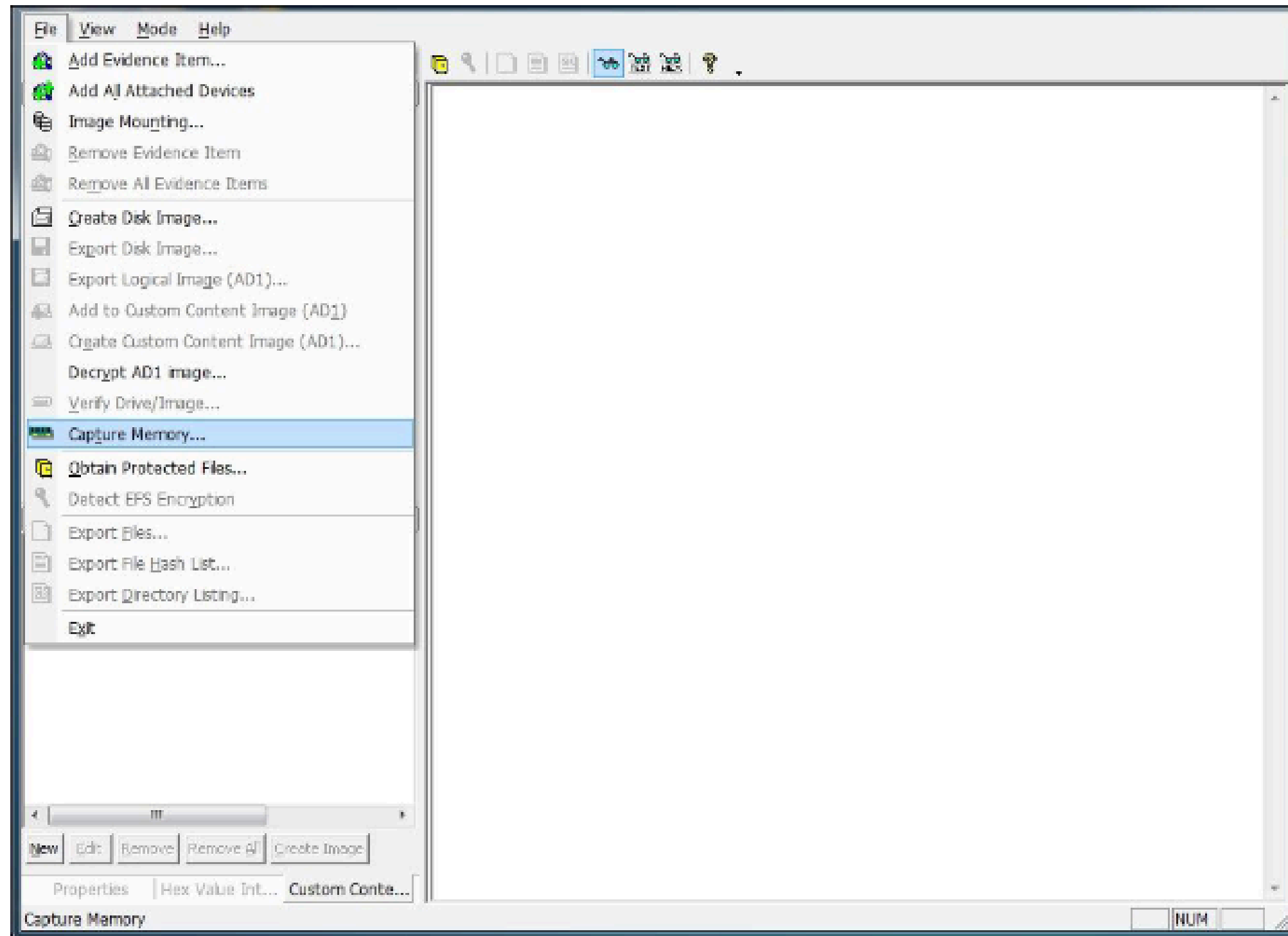
Why Memory Analysis?

- Physical memory is the short-term memory of a computer.
- Physical memory is a storage media like a hard disk drive.
- Physical memory contains unique data, can not be found on other media.
- Some attacks don't leave traces on disk, but only in memory.



Memory Acquisition

- Use FTK Imager to obtain an image for the physical memory.



Memory Analysis: Volatility



What is Volatility:

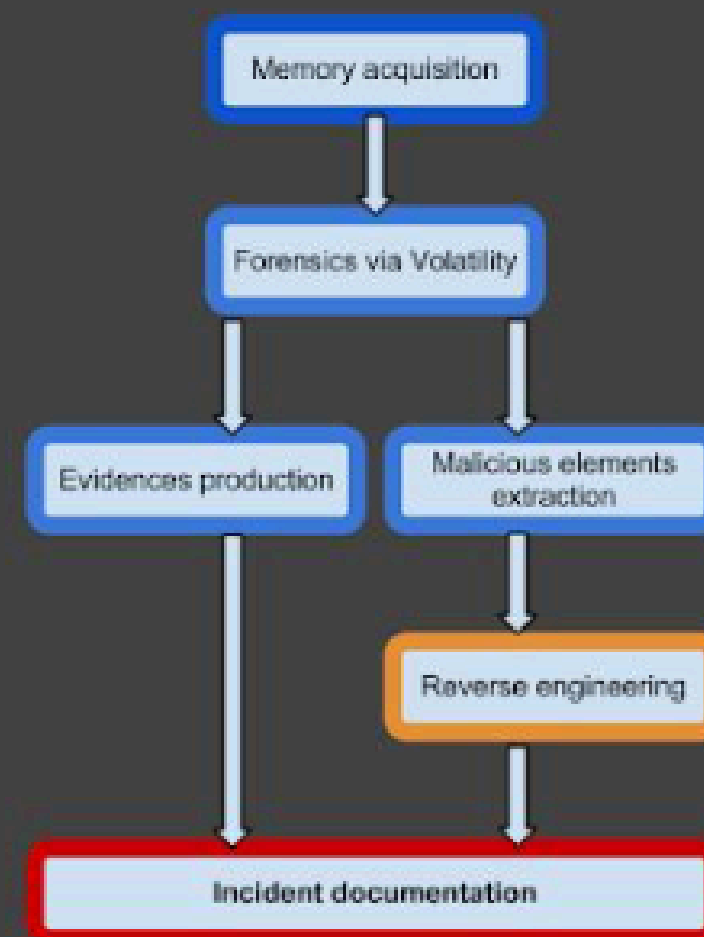
Volatility is a command line tool that allows analysts and forensics investigators to obtain useful information from memory dumps such as processes, network connections, etc.

Installation:

<https://www.volatilityfoundation.org/>

Why volatility?

- Free, Open Source.
- Support X86 X64.
- Windows - Linux - Mac - Android.
- Python extended.



Volatility Usage

- Identify Profile and other Image info: `volatility -f image.dmp imageinfo`
- List all processes: `volatility -f image.dmp -profile=PROFILE pslist`
- List all processes in a tree form: `volatility -f image.dmp pstree`
- List all processes including hidden processes: `volatility -f image.dmp psscan`
- Scan files in physical memory: `volatility -f image.dmp filescan`
- Dump files: `volatility -f image.dmp dumpfiles`
- Shows the contents of the clipboard: `volatility -f image.dmp clipboard`
- Dump an executable from memory: `volatility -f image.dmp procdump`
- Dump SAM hashes: `volatility -f image.dmp hashdump`
- Show network connections: `volatility -f image.dmp netscan`
- Scan the memory for injected code: `volatility -f image.dmp malfind`



Incident Reporting

Type of incident	
Malicious Code	Virus, Trojan, Worm
Attack Status	Running, Closed
Significant	High, Low,...
Number of systems affected	100+

Incident Reporting

INFORMATION SECURITY INCIDENT REPORT

Department: _____

Incident Number: _____
(For Information Security Office only)

A. Notification

1. Date of notification to the Chief Information Security Officer or Departmental Information Security Representative (ISAC member): _____

B. Incident Information

1. Details of Incident:

a) Date incident occurred: _____ ☐ Unknown

b) Date incident detected: _____ ☐ Unknown

c) Incident location: _____

d) General description: _____

e) Media/Device type, if applicable: _____

Was the portable storage device encrypted? ☐ Yes ☐ No

If NO, explain: _____

f) Describe the costs associated with resolving this incident: _____

g) Total estimated cost of incident: _____

2. Incidents involving personally identifiable information

a) Was personally identifiable information involved? ☐ Yes ☐ No (If No, go to Part C)

Type of personally identifiable information (Check all that apply)

- | | |
|---|--|
| ☐ Name | ☐ Health or Medical Information |
| ☐ Social Security Number | ☐ Financial Account Number |
| ☐ Driver's License/State ID Number | |
| ☐ Other (Specify) | |

b) Is a Privacy Disclosure notice required? ☐ Yes ☐ No